

Spatial ML - class 8 - CNN

Total points 100/100 ?

Email *

o.marvan@student.uw.edu.pl

Student index: *

477001

Name: *

Ondřej

Surname: *

Marvan



What is the main output in semantic segmentation? *

10/10

- ☐ A class label for the whole image
- ☒ A class label for each pixel
- ☐ Probability score per chip
- ☐ One regression value per raster

In our workflow, what is the mask? *

10/10

- ☒ The raster of true class labels
- ☐ The validation split
- ☐ The chip index table
- ☐ The RGB image



Why do we use chips? *

10/10

- ☐ To remove class imbalance
- ☒ To split large scenes into manageable training samples
- ☐ To improve image colors
- ☐ To reduce the number of observations

Which architecture is especially suitable for segmentation? *

10/10

- ☐ RNN
- ☒ U-Net
- ☐ XGBoost
- ☐ Random Forest



What is the main role of skip connections in U-Net? * 10/10

- ☐ Speed up data download
- ☐ Decrease class imbalance
- ☐ A different name for decoder
- ☒ Preserve fine spatial boundaries and local detail

Why should train/test split happen by scene, not by random chip? * 10/10

- ☐ To increase chip size
- ☐ It doesn't matter
- ☒ To avoid spatial leakage from neighboring chips
- ☐ To make IoU higher



Which class is usually hardest to segment in land-cover tasks?

*

10/10

- ☐ Water
- ☐ Woodland
- ☐ Background
- ☒ Roads

A model has high pixel accuracy but IoU = 0 for roads. What does this mean?

*10/10

- ☐ The confusion matrix is wrong
- ☐ The model works equally well for all classes
- ☐ The model predicts roads perfectly
- ☒ The model gets large classes right but fails to reconstruct roads



What is the main purpose of augmentation?

*

10/10

- ☐ Increase raster resolution
- ☐ Reduce chip size
- ☒ Expose the model to more visual variability without new labels
- ☐ Remove the need for validation data

Why is IoU often more informative than pixel accuracy?

*

10/10

- ☐ It trains the model faster
- ☐ It replaces the confusion matrix
- ☒ It measures overlap of predicted and true object areas
- ☐ It uses only true positives

This form was created inside of Uniwersytet Warszawski. - [Contact form owner](#)

Does this form look suspicious? [Report](#)



Google Forms



